

4.1 Text Representation: Introduction

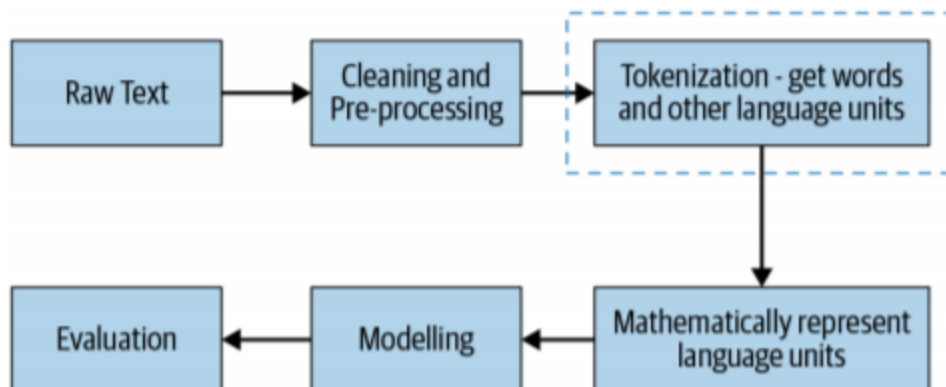
Why models need numbers, and the families of representations.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

1. Why text has to become numbers

A classifier, a ranker, or a neural net does not read English. It reads **vectors**. Week 4 is the map from raw text to those vectors.

The pipeline is always the same shape. You clean the text, split it into units, turn those units into numbers, fit a model, and score the model. This week is the third box: **represent the language units mathematically**.



Images already arrive as numbers (pixel intensities). Text does not. You have to choose a scheme.

Three families show up in this course:

Family	What you store	When you use it
Basic vectorization	Counts or indicators over a vocabulary	Fast baselines, sparse linear models
Distributed representations	Dense vectors learned from context	Similarity, neural models
Handcrafted features	Lengths, lexicons, metadata	Classification when you know the domain

Notes **4.2–4.5** are the basic schemes. Week 5 is the dense ones.

The main text is *Practical Natural Language Processing* (Vajjala, Majumder, Gupta, Surana).

2. What a representation has to do

Suppose the job is sentiment. To get the label right, the system has to get at the meaning of the sentence. A useful representation makes these four things easy to recover:

1. the **lexical units** (words, subwords, phrases)
2. a usable meaning for each unit
3. the **syntax** (who did what to whom)
4. the **context** the sentence sits in

If the scheme cannot support those, the model is guessing from a bag of tokens. That can still work for spam. It fails for anything that depends on order or rare words.

3. A running toy corpus

Every note this week uses the same four documents after lowercasing and stripping punctuation:

Doc	Text
(D_1)	dog bites man
(D_2)	man bites dog
(D_3)	dog eats meat
(D_4)	man eats food

Vocabulary, in first-seen order: [dog, bites, man, eats, meat, food] . Size ($|V| = 6$).

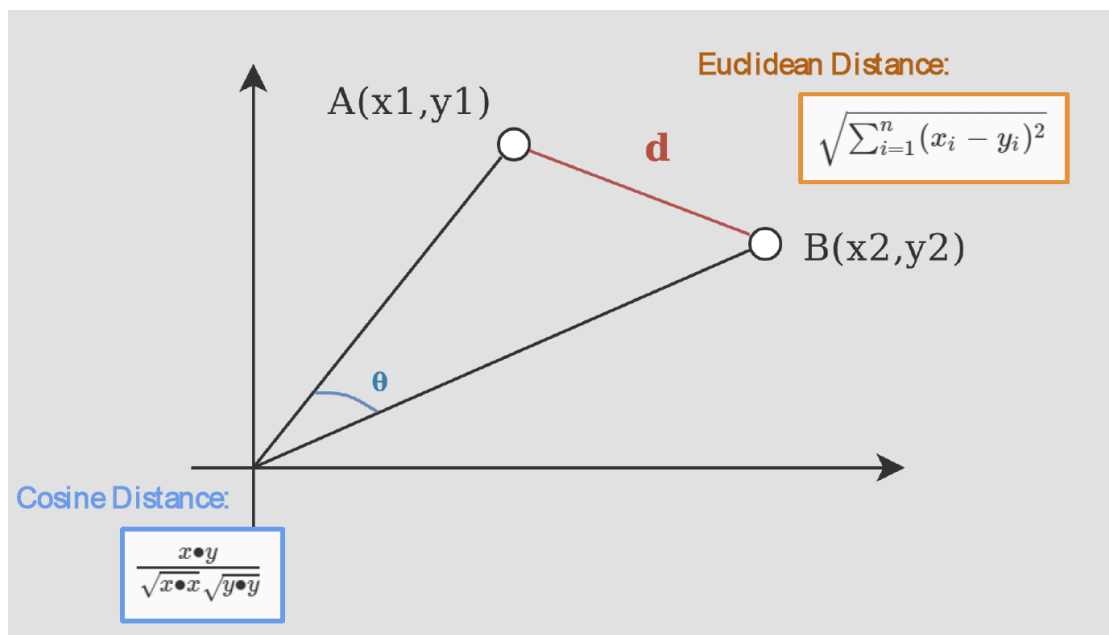
Keep this table. One-hot, bag of words, n-grams, and TF-IDF are all encodings of these four lines.

4. Comparing two documents

Once each document is a vector, you need a number that says how alike two vectors are.

Euclidean distance is the straight-line length between the two tips. It is the “natural” distance, and it blows up when the vectors have different magnitudes. A long review and a short review about the same product look far apart just because one has more counts.

Cosine similarity looks at the **angle** from the origin. ($\cos 0 = 1$). Two documents that use the same words in the same proportions score high even if one is longer. For count vectors, that is usually what you want.



The formula on the cosine side of that figure is similarity, not distance. If you need a distance, use $(1 - \text{cosine similarity})$.

Rule of thumb for this course: cosine for text counts and embeddings; Euclidean only when length itself is a feature you care about.

5. What this week is for

By the end of Week 4 you should be able to:

- say why a model needs a text representation
- write a one-hot matrix and a bag-of-words vector by hand on a tiny corpus
- explain what an n-gram buys you that a unigram bag does not
- compute a TF-IDF weight and say what the IDF term is doing
- pick cosine or Euclidean and defend the choice

Note **4.2** starts with the simplest encoding: one-hot vectors.

6. Practice

1. A 2,000-word essay and a 200-word paragraph use the same words in the same proportions. Why can Euclidean distance call them dissimilar while cosine calls them similar?
2. Name one thing a bag of words cannot recover from (D_1) versus (D_2) .
3. Pixel values are already numbers. What extra job does text representation have to do that an image pipeline does not?